



Teheran University of Medical Sciences

International campus Vice-Dean for Educational Affairs

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Final Exam
International Campus, Pharmacy Program
Biochemistry

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Name and Surname:

Student ID Number:

Course Coordinator: Dr. Panahi

Date of Exam: Dec. 05, 2024

Duration of Exam: 36 minutes

MCQ Question

36

Directions:

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.

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1. Maple syrup disease is a disorder due to a deficiency of pathway that degrades:
 - a) Tyrosin
 - b) Methionine
 - c) Tryptophan
 - d) Isoleucine
2. Which of the following statements about the urea cycle is correct?
 - a) Argininosuccinate is lysed to urea and ornithine in the urea cycle.
 - b) Carbamoyl phosphate supplies both of the nitrogen atoms of urea in the urea cycle.
 - c) Arginine is hydrolysed to urea and ornithine in the urea cycle.
 - d) The formation of urea from the urea cycle yields energy.
3. Which of the following statements about transamination reactions is correct?
 - a) Transamination reactions involve ATP hydrolysis.
 - b) Transamination reactions are irreversible.
 - c) Transamination reactions require NAD^+ or NADP^+ .
 - d) Transamination reactions require pyridoxal-5'-phosphate (PLP).
4. Which of the following amino acids cannot provide a substrate for gluconeogenesis?
 - a) Leucine
 - b) Tryptophan
 - c) Histidine
 - d) Isoleucine
5. Amino acids are released into the portal system by:
 - a) Pinocytosis
 - b) Active transport
 - c) Facilitated diffusion
 - d) Passive transport



6. Which of the following enzyme is deficient in alkaptonuria?

- a) Homogentisate dioxygenase
- ☒ b) p-Hydroxyphenyl- pyruvate dioxygenase
- c) Phenylalanine hydroxylase
- d) Tyrosinase

7. Which of the following enzyme catalyze the major mechanism for transporting ammonia in a nontoxic form?

- a) Aminotransferase
- b) Amino acid oxidase
- ☒ c) Glutamine synthetase
- d) Threonine dehydrogenase

8. Identify the amino acid that participates in methyl group transfer:

- a) Lysine
- b) Serine
- c) Valine
- ☒ d) Methionine

9. Which of the following is the only plasma enzyme normally found in urine?

- a) Acid Phosphatase
- b) Amylase
- ☒ c) Nucleotidase
- d) Aminotransferase

10. Which of the following is the most important cause of increased aminotransaminase activity in serum?

- ☒ a) Bone disease
- b) Liver disease
- c) Pancreatitis
- d) Brain disease

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um nonfunctional enzymes
Alkaline phosphatase
Acid phosphatase
Lipoprotein
Gamma

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11. Serum nonfunctional enzymes are all, except:

- a) Alkaline phosphatase
- b) Acid phosphatase
- c) Lipoprotein lipase
- d) Gamma glutamyltranspeptidase

12. What enzyme in the body is measured to diagnose rhabdomyolysis?

- a) Lipase
- b) Nucleotidase
- c) Creatine kinase
- d) Glutamyltransferase

13- All the following compounds are involved in the biosynthesis of purine nucleotides except:

- A) Asparagine
- B) phosphoribosyl pyrophosphate
- C) Glutamine
- D) Glycine

14- In which of the following diseases conjugated bilirubin increases?

- A) Gilbert's syndrome
- b) Dubin-Johnson syndrome
- C) Carpenter Crigler syndrome type 2
- d) Hemolytic anemia



15- Allopurinol inhibits which of the following enzymes?

- A) dihydrofolate reductase
- b) adenosine deaminase
- c) Xanthine oxidase
- D) ribonucleotide reductase

16- What is the end product of urobilinogen metabolism in the kidney?

- A. Urobilinogen
- B. Urobilin
- C. Stercobilin
- D. Bilirubin
- E.

17- Creatine is synthesized from which of the following compounds in the liver and kidneys?

- A) Glucose
- B) Arginine
- C) Lysine
- D) phenylalanine

18- Fluorouracil inhibits which of the following enzymes?

- A) Ribonucleotide Reductase
- B) Thymidylate Synthase
- C) Hypoxanthine Guanine Phosphoribosyl Transferase
- D) Xanthine Oxidase

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Which of the following is an inhibitor of mitochondrial inner membrane?
A) Rotenone
B) Cyanide
C) Atractylis

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19- Which of the following is an inhibitor of ATP/ADP transporter in the mitochondrial inner membrane?

- A) Oligomycin ✓
- b) CN⁻
- c) Atractyloside
- d) Barbiturates

20- Which part of the respiratory chain has the lowest tendency to oxidize?

- A) Flavin Mononucleotide
- b) Cytochrome Oxidase
- C) Cytochrome c
- d) Succinate Thiokinase

21- Which of the following is an uncoupler of oxidation from phosphorylation?

- A) Thermogenin ✓
- B) Carboxin
- C) Malonate
- D) Rotenone

22- In the conversion of pyruvic acid to acetyl coenzyme-A, all of the following are involved, Except:

- A. FAD
- B. TPP
- C. CoA
- D. Biotin



23- All of the following factors play a role in chylomicron metabolism, except:

- A. LPL (lipoprotein lipase)
- ~~B. CETP (cholesteryl ester transfer protein)~~
- C. HDL-C (high-density lipoprotein cholesterol)
- D. Apo CII (apolipoprotein CII)

24- Which of the following activates the enzyme HMG-CoA reductase?

- A. cAMP
- B. Atorvastatin
- ~~C. Insulin~~
- D. Protein Kinase

25- From the beta-oxidation of myristic acid in the mitochondria, all of the following compounds are produced, except:

- A. Malonyl CoA
- B. Acetyl CoA
- ~~C. FADH₂~~
- D. NADH

26- Which compound is involved in the transfer of acetyl-CoA from the mitochondria to the cytosol?

- A. Malate
- ~~B. Carnitine~~
- C. Pyruvate
- D. Citrate

27- For the synthesis of palmitic acid, all of the following are required, except:

- A. Carnitine
- B. Acetyl-CoA
- ~~C. ATP~~
- D. NADPH+H

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28- An enzyme that is essential for both ketone body synthesis and cholesterol synthesis is:

- A. Acetyl CoA carboxylase
- ☒ B. HMG CoA synthase
- C. HMG CoA reductase
- D. HMG CoA lyase

29- Type I hyperlipoproteinemia is characterized by:

- A. Elevated LDL levels
- B. Elevated Cholesterol levels
- ☒ C. Elevated chylomicron levels
- D. Elevated HDL levels

30. Net ATP generation on complete oxidation of stearic acid is?

- ☒ A. 129
- B. 131
- C. 146
- D. 148

31- Which of the following apolipoproteins activates LCAT?

- A. C-II
- B. A-II
- C. B-48
- ☒ D. A-I

32. Which of the following is the rate-limiting step of fatty acid oxidation and is also inhibited by malonyl CoA?

- A. Thiokinase
- B. Carnitine Palmitoyl transferase I
- ☒ C. Acyl CoA Dehydrogenase
- D. Thiolase



33- Which one of the following enzymes is involved in the conversion of IDL to LDL?

- ☒ A. Hepatic lipase
- ☐ B. Lipoprotein lipase
- ☐ C. Hormone-sensitive lipase
- ☐ D. Pancreatic lipase

34. Lesch-Nyhan syndrome, which is an X-linked recessive disorder, is caused by a deficiency of which enzyme?

- ☐ A. Hypoxanthine-Guanine Phosphoribosyl Transferase
- ☒ B. Adenosine Deaminase
- ☐ C. Xanthine Oxidase
- ☐ D. Adenine Phosphoribosyl Transferase

35. Porphyrria cutanea tarda is caused by a deficiency in which one of the following enzymes?

- ☐ A. ALA synthase
- ☐ B. Ferrochelatase
- ☒ C. Uroporphyrinogen decarboxylase
- ☐ D. Coproporphyrinogen oxidase

36. Where are FAD and Cytochrome aa₃ located in the electron transport chain?

- ☐ A. I and III
- ☐ B. II and IV
- ☐ C. II and III
- ☒ D. III and IV